TT200 Heraeus

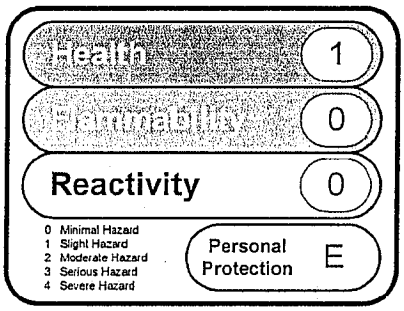
MATERIAL SAFETY DATA SHEET

**TECTIP®**



## 1. PRODUCT and COMPANY IDENTIFICATION

**Heraeus Electro-Nite Co.**
1 Summit Square, Suite 100
Langhorne, PA 19047




**Emergency Telephone Number:** (215) 944-9000

**Product Name:** TECTIP®

**Product Use:** TECTIP® is a single use device with a thermocouple for the measurement phase changes in molten iron

## 2. HAZARDS IDENTIFICATION

**Eyes**
Dust or particles can cause mild eye irritation.

**Skin**
Dust may cause mild skin irritation.

**Ingestion**
Ingestion of any of the ingredients of this product is a health risk.

**Inhalation**
Excessive inhalation may cause irritation to the nose, throat, and lungs.

This product classifies under OSHA as an "article" and therefore does not require a material safety data sheet. These sheets are produced and provided at the request of the consumer. OSHA defines an "article" as a manufactured item: (1) which is formed to a specific shape or design during manufacture; (2) which has end use function(s) dependent in whole or in part upon its shape or design during end use; and (3) which does not release, or otherwise result in exposure to, a hazardous chemical under normal conditions of use.

## 3. COMPOSITION and INGREDIENT INFORMATION

| Significant Ingredient | CAS Number | Weight % | Exposure Limits ($mg/m^3$) OSHA PEL | ACGIH TLV |
|---|---|---|---|---|
| Silica (Quartz) | 14808-60-7 | 92-96 | 2.1 Dust | 0.1 Dust |
| Calcium Aluminate Mix | 12042-78-3 | 2-4 | 5 Dust | 10 Dust |
| Ceramic | 94552-04-2 | 1-2 | 1 Dust | 1 Dust |
| Tellurium | 13494-80-9 | 0-1.5 | 0.1 | 0.1 |

While this material is not considered hazardous by the OSHA Hazard communication Standard (29 CFR 1910.1200), this Material Safety Data Sheet contains valuable information on the safe handling and proper use of the material.

Electro-Nite

## 4. FIRST AID MEASURES

**Eyes**
Flush our dust or particles for 15 minutes in a gentle stream of water. Consult physician if irritation persists.

**Skin**
Wash off dust with soap and water. Consult physician if irritation persists.

**Ingestion**
If any ingredients are swallowed, consult a physician.

**Inhalation**
Remove to fresh air. Immediately consult physician for any breathing difficulty.

## 5. FIRE FIGHTING MEASURES

**Flammability Properties**
Flammability properties such as flash point, LFL, and UFL are not applicable for this product. Cellulose fiber tube will burn if exposed to sufficient heat.

**Extinction Media**
Water, Dry, Chemical. Use extinction media appropriate to the surrounding fire.

**Combustion Products**
Carbon Monoxide, Carbon Dioxide.

**Firefighting Instructions**
Wear self-contained breathing apparatus and protective suit.

## 6. ACCIDENTAL RELEASE MEASURES

**Personal Precautions**
See personal protection information in Section 8.

**Environmental Precautions**
Keep extinguishing water away from drains, surface and ground water, and soil.

**Methods for Clean-up**
Manual.

## 7. HANDLING and STORAGE

**Handling**
Wear appropriate protective equipment and follow all company safety procedures.

**Storage**
Store in a cool, dry, well-ventilated area that is away from molten metal splashing.

## 8. EXPOSURE CONTROLS and PERSONAL PROTECTION

**Engineering Controls**
Area must be well ventilated.

**Eye/Face Protection**
Safety glasses, hard hat, and face shield are required for safe furnace work.

**Skin Protection**
Fire retardant clothing and gloves, as well as safety shoes are required for safe furnace work.

**Respiratory Protection**
Some furnace operators may require the use of a NIOSH approved dust or fume respirator.

**Hygiene Considerations**
Use safe furnace practices when using this product.

## 9. PHYSICAL and CHEMICAL PROPERTIES

**Appearance**
Solid. Brownish-yellow square cup, inside is a thermocouple wire enclosed in a thin glass tube with a tellurium deposit inside the base of the cup.

**Color/Odor**
Not applicable.

**Physical Properties**
Physical constants such as boiling point, evaporation rate, and pH are not available for this product.

## 10. STABILITY and REACTIVITY

**Stability**
Stable.

**Conditions to Avoid**
Incompatible with Hydrofluoric acid.

**Hazardous Decomposition Products**
Does not decompose, stable under normal conditions.

## 11. TOXICOLOGY INFORMATION

**Acute Effects**
Tellurium has an LD50 of 83 mg/kg in rats.

**Chronic Effects / Carcinogenicity**
Breathing of silica for many years can cause lung disease, however most silica in this product is not airborne. Limited and contradictory evidence suggest that silica may be a carcinogen when inhaled for many years (IARC).

## 12. ECOLOGICAL INFORMATION

**Ecotoxicity**
Keep extinguishing water away from drains, surface and ground water, and soil.

**Degradability**
Not applicable.

**Bioaccumulation**
None.

## 13. DISPOSAL CONSIDERATIONS

**Disposal**
After use, discard product in furnace or slag pit. When discarded this product is not considered a hazardous waste.

## 14. TRANSPORTATION INFORMATION

**Transportation**
This product is not subject to regulations on transport and labeling that apply to dangerous substances.

## 15. REGULATORY INFORMATION

**Disclaimer:** This section is not intended to be a comprehensive list of regulations that may apply to this product.

**U.S. Occupational Safety and Health Administration**
This product classifies under OSHA as an "article" and therefore does not require a material safety data sheet. OSHA defines an "article" as a manufactured item: (1) which is formed to a specific shape or design during manufacture; (2) which has end use function(s) dependent in whole or in part upon its shape or design during end use; and (3) which does not release, or otherwise result in exposure to, a hazardous chemical under normal conditions of use (29 CFR 1910.1200(c)).

**California Proposition 65**
This product is not known to contain any chemicals listed as carcinogens or reproductive toxins under California Proposition 65 at levels which would be subject to proposition.

**WHMIS Canadian Regulation**
This product has been classified in accordance with the hazard criteria of the Controlled Products Regulations and the MSDS contains all the information required by the Controlled Products Regulation.

## 16. OTHER INFORMATION

| | | |
|---|---|---|
| **HMIS Hazard Rating:** | Health | 1 |
| | Fire | 0 |
| | Reactivity | 0 |
| | Personal Protection | E |

**Revision Number:** 3
**Revision Date:** July 1, 2006

**Supersedes Date:** March 30, 2004

**HMIS Personal Protection**

| | |
|---|---|
| A | Safety glasses |
| B | Safety glasses and gloves |
| C | Safety glasses, gloves and an apron |
| D | Face shield, gloves and an apron |
| E | Safety glasses, gloves and a dust respirator |
| F | Safety glasses, gloves, apron and a dust respirator |
| G | Safety glasses, a vapor respirator |
| H | Splash goggles, gloves, apron and a vapor respirator |
| I | Safety glasses, gloves and a dust/vapor respirator |
| J | Splash goggles, gloves, apron and a dust/vapor respirator |
| K | Airline hood or mask, gloves, full suit and boots |
| L - Z | Custom PPE specified by employer |

**Acronyms**

| | |
|---|---|
| ACGIH | American Conference of Governmental Industrial Hygienists |
| CAS | Chemical Abstracts Service |
| CFR | Code of Federal Regulations |
| HMIS | Hazardous Material Information System |
| IARC | International Agency for Research on Cancer |
| NIOSH | National Institute of Occupational Safety and Health |
| OSHA | U.S. Occupational Safety & Health Administration |
| PEL | Permissible Exposure Limit |
| TLV | Threshold Limit Value |
| WHMIS | Workplace Hazardous Materials Information System |

**Disclaimer of expressed and implied warranties:**

This information is given in good faith and is taken from sources believed reliable and accurate as of the date stated. No representation or warranties either expressed or implied of merchantability, fitness for a particular purpose or of any other nature are made hereunder with respect to the information set forth herein or to the product to which the information refers. This product should be used only as directed with proper safety equipment and clothing. Safety equipment, including hard hat, safety glasses, gloves and safety shoes should be used. Face shields, fire-retardant clothing, metatarsal guards and leggings should be used when required. Contact your supervisor for safety requirements and operating procedure.